Matrix Multiplication

Let $A = (a_{ij})_{m\times p}$, $B = (b_{ij})_{p\times n}$.

The matrix product AB is given by:

$$(AB)_{ij} = \sum_{k=1}^{p} \underset{\substack{\uparrow \\ \text{ith row} \\ \text{of } A}}{a_{ik}} \underset{\substack{\uparrow \\ \text{jth column} \\ \text{of } B}}{b_{kj}}.$$

This is just a dot product:

$$(AB)_{ij} = (\text{ith row of } A) \cdot (\text{jth col of } B).$$

⚠ In general, $AB \neq BA$.

Ex. Let $A = \begin{pmatrix} -1 & 0 \\ 2 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix}$.

We can compute products:

$$AB = \begin{pmatrix} -1 & -2 \\ 11 & 4 \end{pmatrix}, \quad BA = \begin{pmatrix} 3 & 6 \\ -3 & 0 \end{pmatrix}$$

So, $AB \neq BA$.

Properties. If A is $m\times p$, B is $p\times n$, and for a matrix C of appropriate size:

- $A(BC) = (AB)C = ABC$
- If B and C are the same size:
  $A(B+C) = AB + AC$
- $k(AB) = (kA)B = A(kB)$ for $k \in \mathbb{R}$

⚠ If $AB = 0$, A and B may not be 0!

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

We can express linear systems as matrix products:

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \qquad\qquad \vdots \qquad \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

$$\Rightarrow \underset{\substack{m\times n \\ A}}{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}} \underset{\substack{n\times 1 \\ \vec{x}}}{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}} = \underset{\substack{m\times 1 \\ \vec{b}}}{\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}}$$

Vectors in Euclidean Space

Given a linear system, we can express it as a vector equation:

$$x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

An n-vector is a sequence of n numbers:

$$\vec{v} = (v_1, v_2, \ldots, v_n).$$

The set of all n-vectors is called Euclidean n-space, $\mathbb{R}^n$. We can associate $\vec{v}$ with the $n \times 1$ column matrix:

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

If $\vec{v}, \vec{w} \in \mathbb{R}^n$:

- $\vec{v} = \vec{w}$ if and only if $v_i = w_i$ for all i
- $\vec{v} + \vec{w} = (v_1 + w_1, \ldots, v_n + w_n) \in \mathbb{R}^n$
- $k\vec{v} = (kv_1, \ldots, kv_n) \in \mathbb{R}^n$

We say that a vector $\vec{v} \in \mathbb{R}^n$ is a linear combination of $\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k \in \mathbb{R}^n$ if

$$\vec{v} = c_1\vec{u}_1 + c_2\vec{u}_2 + \cdots + c_k\vec{u}_k$$

for some $c_1, c_2, \ldots, c_k \in \mathbb{R}$.

Ex. Is $(3,4)$ a linear combination of $(5,2)$ and $(1,-1)$? Yes!

This is just asking if we can write

$$x \begin{pmatrix} 5 \\ 2 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

for some $x, y \in \mathbb{R}$. Indeed we can: take $x = 1$, $y = -2$.

Ex. · Is $(9,2,7)$ a linear combination of $(1,2,-1)$ and $(6,4,2)$?

· Is $(4,-1,8)$ a linear combination of $(1,2,-1)$ and $(6,4,2)$?

Ex: · Is $(9,2,7)$ a linear combination of $(1,2,-1)$ and $(6,4,2)$?

· Is $(4,-1,8)$ a linear combination of $(1,2,-1)$ and $(6,4,2)$? In $\mathbb{R}^3$.

We check:

$$x\begin{pmatrix}1\\2\\-1\end{pmatrix} + y\begin{pmatrix}6\\4\\2\end{pmatrix} = \begin{pmatrix}9\\2\\7\end{pmatrix}\ ?$$

$$\Rightarrow \begin{cases} x + 6y = 9 \\ 2x + 4y = 2 \\ -x + 2y = 7 \end{cases}$$

$$\left(\begin{array}{cc|c} 1 & 6 & 9 \\ 2 & 4 & 2 \\ -1 & 2 & 7 \end{array}\right) \xrightarrow[R_3+R_1]{R_2-2R_1} \left(\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & -8 & -16 \\ 0 & 8 & 16 \end{array}\right) \xrightarrow{R_3+R_2} \left(\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & -8 & -16 \\ 0 & 0 & 0 \end{array}\right)$$

The linear system is consistent!

Yes, $(9,2,7)$ is a linear combination of $(1,2,-1)$ and $(6,4,2)$.

We check:

$$x\begin{pmatrix}1\\2\\-1\end{pmatrix} + y\begin{pmatrix}6\\4\\2\end{pmatrix} = \begin{pmatrix}4\\-1\\8\end{pmatrix}\ ?$$

$$\left(\begin{array}{cc|c} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 8 \end{array}\right) \xrightarrow[R_3+R_1]{R_2-2R_1} \left(\begin{array}{cc|c} 1 & 6 & 4 \\ 0 & -8 & -9 \\ 0 & 8 & 12 \end{array}\right) \xrightarrow{R_3+R_2} \left(\begin{array}{cc|c} 1 & 6 & 9 \\ 0 & -8 & -16 \\ 0 & 0 & 3 \end{array}\right)$$

$0 = 3$

The linear system is inconsistent!

No, $(4,-1,8)$ is NOT a linear combination.

Geometry

In $\mathbb{R}^2$, a line is given by:

- algebraic equation $ax + by = c$
- solving, we have

$$x = \frac{c-by}{a}.$$

Letting $y = t$, we have parametric eqs.

$$x = \frac{c-bt}{a}, \quad y = t, \quad t \in \mathbb{R}.$$

- as a vector equation:

$$\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}(c-bt)/a\\t\end{pmatrix} = \begin{pmatrix}c/a\\0\end{pmatrix} + \begin{pmatrix}-b/a\\1\end{pmatrix}t, \quad t \in \mathbb{R}.$$

* We also have lines in $\mathbb{R}^3$:

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}x_0\\y_0\\z_0\end{pmatrix} + t\begin{pmatrix}a\\b\\c\end{pmatrix}, \quad t \in \mathbb{R}.$$

In $\mathbb{R}^n$, lines are given by:

$$\vec{x} = \vec{x}_0 + t\vec{v}, \quad t \in \mathbb{R}.$$

This is a line through $\vec{x}_0$ parallel to $\vec{v}$.

In $\mathbb{R}^3$, a plane is given by:

- algebraic equation: $ax + by + cz = d$
- parametric equations: let $y = s$, $z = t$

$$x = \frac{d - bs - ct}{a}, \quad y = s, \quad z = t, \quad s, t \in \mathbb{R}.$$

- vector equation:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} d/a \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -b/a \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -c/a \\ 0 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

* In $\mathbb{R}^n$, planes are given by

$$\vec{x} = \vec{x}_0 + s\vec{v} + t\vec{w} \quad \text{for } s, t \in \mathbb{R},$$

where $\vec{v} \neq k\vec{w}$. This is the plane through $\vec{x}_0$ and parallel to both $\vec{v}$, $\vec{w}$.